Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**HCPA**
Semester	**5th**	**Course Code**	**23ADS1524**
Roll No	**119**	**Name of Student**	**Shrusti Katakwar**

Practical Number	**01**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.

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Aim	**Write OpenMP programs for parallelizing for loops with different scheduling strategies.**
Problem Definition	**Write OpenMP programs for parallelizing for loops with different scheduling strategies.**
Theory (100 words)	OpenMP is an API used for parallel programming in shared-memory systems. It allows a loop to be divided among multiple threads so that iterations can execute simultaneously. For parallelizing a for loop, OpenMP provides the parallel for directive. The scheduling strategy determines how loop iterations are distributed among the available threads. **1. Static Scheduling** In static scheduling, loop iterations are divided among threads before execution starts. ● Each thread receives a predetermined set of iterations. ● It has very low scheduling overhead. ● It works best when all iterations require approximately the same amount of computation. ● A chunk size can be specified to control the number of iterations assigned at a time. Example: If there are 20 iterations and 4 threads, the iterations can be divided approximately equally among the four threads. Advantage: Low overhead and predictable execution. Disadvantage: Poor load balancing if some iterations take much longer than others. --- **2. Dynamic Scheduling** In dynamic scheduling, loop iterations are assigned to threads during execution. ● A thread receives a chunk of iterations. ● After completing its assigned chunk, it requests another available chunk. ● This continues until all iterations are completed. ● It is useful when different iterations require different amounts of computation. Advantage: Provides good load balancing. Disadvantage: Higher scheduling overhead compared with static scheduling.

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3. Guided Scheduling

Guided scheduling is a dynamic scheduling method in which the size of the chunks decreases as the loop progresses.

- Initially, threads receive relatively large chunks.
- As the number of remaining iterations decreases, the chunk size becomes smaller.
- A minimum chunk size can be specified.
- It is useful for loops where the workload is irregular.

Advantage: Good load balancing with generally less overhead than fine-grained dynamic scheduling.
Disadvantage: More complex than static scheduling and may still have scheduling overhead.

4. Runtime Scheduling

In runtime scheduling, the scheduling strategy is not fixed directly in the program.

Instead, the scheduling policy is selected at program execution time, typically using the OMP_SCHEDULE environment variable.

For example, the user can choose static, dynamic, or guided scheduling when running the program.

Advantage: Allows easy experimentation and performance tuning without changing or recompiling the program.
Disadvantage: The actual scheduling behavior depends on the runtime configuration.

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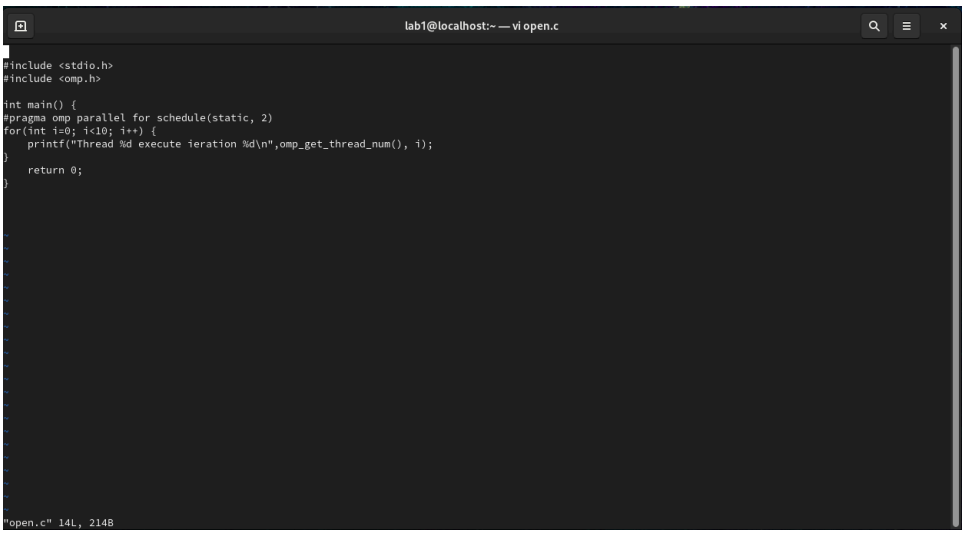
Procedure and Execution

(100 Words)

Step for Implementation:

1. Include OpenMP library in the program.
2. Define the number of loop iterations.
3. Set the number of threads to be used.
4. Apply the parallel for directive to parallelize the loop.
5. Select a scheduling strategy:
   - Static
   - Dynamic
   - Guided
   - Runtime
6. Specify the chunk size if required.
7. Execute the loop using multiple threads.
8. Display the thread number and iteration number to observe how work is distributed.
9. Compile the program using the OpenMP compiler option.
10. Run the program and compare the different scheduling strategies.

Code:

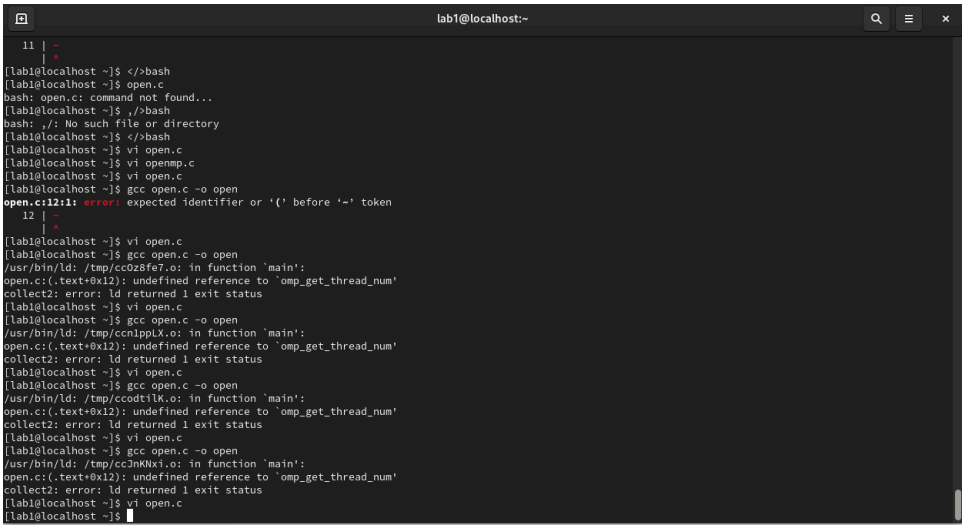


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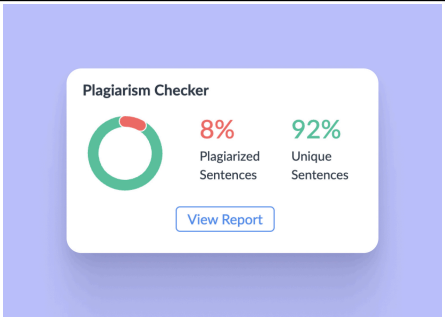
	Output: 
Output Analysis	• Static Scheduling: Loop iterations are divided among threads in a fixed manner. The workload distribution is generally predictable. • Dynamic Scheduling: Threads receive new iterations as they finish their previous work, giving better load balancing. • Guided Scheduling: Larger chunks are assigned initially, followed by smaller chunks. This balances workload while reducing scheduling overhead. • Runtime Scheduling: The output depends on the scheduling strategy selected at execution time. • The thread number shown in the output indicates which thread executed each iteration. • The order of output may vary between executions because multiple threads run simultaneously. Result: Different scheduling strategies distribute loop iterations differently, and the suitable strategy depends on the workload and required load balancing.
Link of student Github profile where lab assignment has been uploaded	

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Conclusion	OpenMP successfully parallelizes for loops by distributing iterations among multiple threads. Different scheduling strategies such as static, dynamic, guided, and runtime provide different methods of workload distribution. Proper selection of a scheduling strategy improves load balancing, execution efficiency, and overall parallel performance.
Plag Report (Similarity index < 12%)	
Date	**19/08/26**